

A 10-bit Charge-Redistribution ADC Consuming 1.9 W at 1 MS/s

A Project Report Submitted

in Partial Fulfilment of the Requirements

for the Degree of

Master of Technology

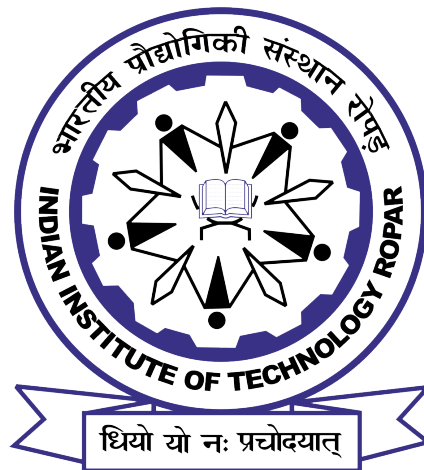
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13-April-2026

Declaration

I **Santosh Kumar Verma** declares that this project report titled as “**A 10-bit Charge-Redistribution ADC Consuming 1.9 W at 1 MS/s**” is my own work carried out under the supervision of **Prof. Devarshi Mrinal Das**. All sources of information used in this report have been properly acknowledged. I confirm that the contents of this report have been prepared with honesty and integrity and do not involve any form of falsification, misrepresentation or unfair means .

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Acknowledgement

I would like to express my sincere gratitude to my supervisor **Prof. Devarshi Mrinal Das** for their guidance and support throughout this project. I am also thankful to the Department of Electrical Engineering, Indian Institute of Technology Ropar, for providing the necessary resources. Finally, I am grateful to my family and friends for their constant encouragement.

Santosh Kumar Verma

Certificate

This is to certify that the project report titled “**A 10-bit Charge-Redistribution ADC Consuming 1.9 W at 1 MS/s** ” submitted by **Santosh Kumar Verma** is a record of work carried out under my supervision. To the best of my knowledge, this work is original and has not been submitted elsewhere for the award of any degree . The report meets the requirements for the award of the Master of Technology .

Signature of the Supervisor
Prof. Devarshi Mrinal Das

Abstract

This paper presents the design of a 10-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) implemented in 65 nm CMOS technology. The proposed architecture focuses on achieving low power consumption by employing a charge-redistribution digital-to-analog converter (DAC) with step-wise charging, along with a dynamic comparator and a delay-line-based control scheme.

The ADC operates with a single supply voltage in the range of 1.0 V to 1.3 V and does not require an external reference current. Its power consumption is mainly governed by the sampling frequency due to its dynamic operation. The layout occupies an approximate area of $115 \times 225 \mu\text{m}^2$. At a sampling rate of 1 MS/s and a supply voltage of 1.0 V, the ADC consumes $1.9 \mu\text{W}$ of power while achieving an energy efficiency of 4.4 J per conversion step. These features make the design well-suited for low-power and energy-efficient applications.

Key Words- Analog-to-Digital Converter (ADC), Successive Approximation Register (SAR), Charge Redistribution, CMOS Technology, Dynamic Comparator, Digital-to-Analog Converter (DAC), Low-Power Design, Energy Efficiency, Wireless Sensor Networks.

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Chapter 1

Introduction

1.1 Overview of Analog to Digital Converters

An Analog-to-Digital Converter (ADC) is a key component in modern electronic systems that converts continuous analog signals into discrete digital values. Since most real-world signals, such as temperature, sound, and voltage, are analog in nature, ADCs are required to make these signals compatible with digital processing systems.

With the rapid growth of portable devices, wireless communication, and sensor-based applications, there is an increasing demand for ADCs that operate with low power while maintaining reliable performance. This is especially important in applications such as Internet of Things (IoT) devices and wireless sensor networks, where energy efficiency is a critical requirement.

The performance of an ADC is primarily determined by its speed, power consumption, resolution, and accuracy. With the advancement of VLSI technology, there is an increasing demand for high-speed and low-power ADCs in portable and battery-operated devices.

1.2 Role of Comparator in ADC

A comparator is one of the fundamental building blocks in most ADC architectures, especially in high-speed converters such as flash ADCs. It is responsible for comparing an input signal with a reference signal and producing a digital output based on the comparison.

The speed and accuracy of the comparator directly affect the overall performance of the ADC. Therefore, designing efficient comparators is critical for achieving high-speed and low-power ADC operation.

1.3 Need for Low-Power High-Speed Comparators

In modern CMOS technologies, supply voltages are continuously scaled down to reduce power consumption. However, this reduction in supply voltage makes the design of high-speed comparators more challenging.

Lower supply voltages reduce the available voltage headroom and transconductance of transistors, which leads to increased delay and degraded performance. Additionally, achieving both high speed and low power simultaneously requires careful design trade-offs.

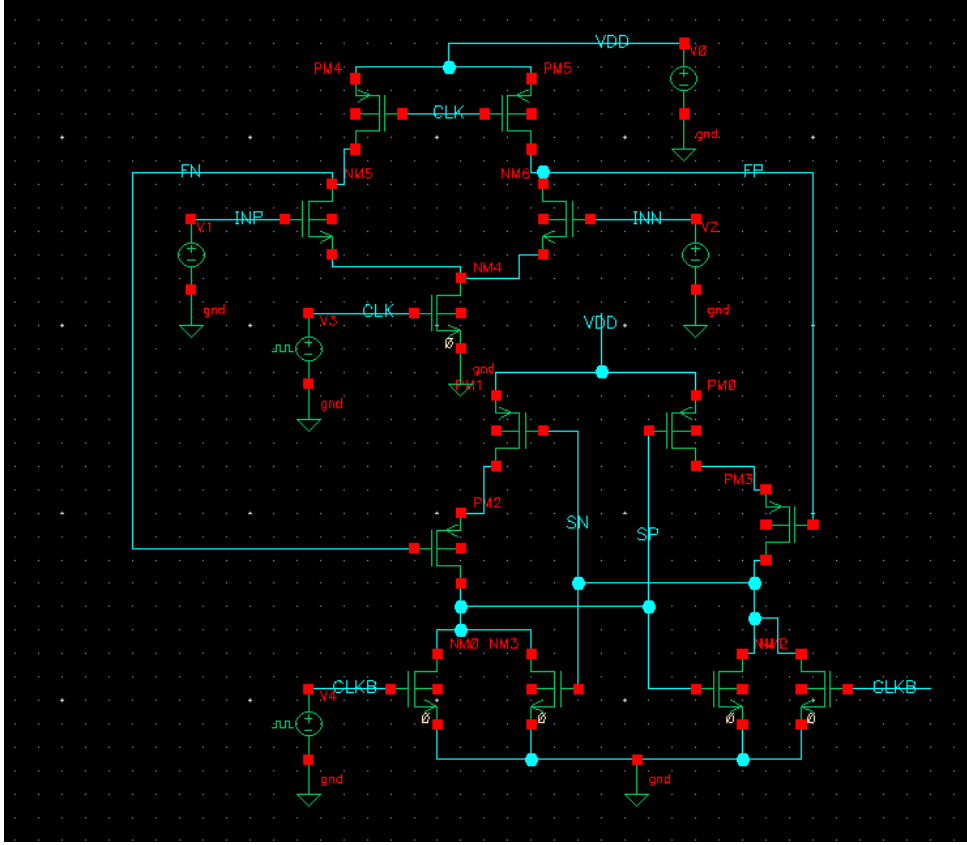


Figure 1.1: Schematic diagram of Energy-efficient dynamic two-stage comparator

Dynamic regenerative comparators are widely used to address these challenges due to their high speed and zero static power consumption.

1.4 Problem Statement

Despite the advantages of conventional dynamic comparators, they suffer from several limitations, especially at low supply voltages. The presence of stacked transistors increases delay and reduces performance.

Although double-tail comparators improve low-voltage operation, they still exhibit limitations in terms of delay and power consumption. Therefore, there is a need for improved comparator architectures that can operate efficiently at low supply voltages while maintaining high speed and low power consumption.

1.5 Objective of the Work

The main objective of this work is to design and analyze a low-power Analog-to-Digital Converter (ADC) based on the Successive Approximation Register (SAR) architecture. The focus is on achieving efficient conversion with reduced power consumption while maintaining acceptable accuracy and speed.

This work aims to implement a charge-redistribution digital-to-analog converter (DAC) that improves energy efficiency through optimized capacitor switching techniques. Additionally, a dynamic comparator is considered to enable fast and reliable decision-making during the conversion process.

Another objective is to develop an effective control mechanism that ensures proper timing and coordination of the conversion steps. The overall goal is to enhance the performance of the ADC in terms of power efficiency, area utilization, and conversion accuracy, making it suitable for low-power and portable electronic applications.

1.5.1 Input-Referred Noise Analysis of the Comparator

The accuracy and efficiency of the comparator are largely influenced by the noise generated in its first stage. Since this stage performs initial amplification, its noise contribution dominates the overall input-referred noise of the system. The primary noise source in this stage is the thermal noise of the input transistors.

The thermal noise of a MOS transistor can be represented by an equivalent resistance at its input, given by

$$R_n \approx \frac{\gamma}{g_m} \quad (1.1)$$

where γ is the noise coefficient (typically close to 1) and g_m is the transconductance of the transistor.

For a differential input pair, the combined noise resistance can be approximated as

$$R_{n,\text{eq}} \approx \frac{2}{g_m} \quad (1.2)$$

During operation, the comparator integrates both signal and noise over a finite time interval. The resulting input-referred noise voltage is expressed as

$$\sigma_v \approx \sqrt{4kT \cdot R_{n,\text{eq}} \cdot B_n} \quad (1.3)$$

where k is Boltzmann's constant, T is absolute temperature, and B_n represents the effective noise bandwidth.

For a finite integration time T_{int} , the noise bandwidth is approximated as

$$B_n = \frac{1}{2T_{\text{int}}} \quad (1.4)$$

The integration time depends on the charging behavior of parasitic capacitances at the output nodes and can be written as

$$T_{\text{int}} \approx \frac{V_{\text{th}} \cdot C}{I} \quad (1.5)$$

where V_{th} is the threshold voltage, C is the effective capacitance, and I is the bias current.

Substituting (2), (4), and (5) into (3), the noise expression becomes

$$\sigma_v \approx \sqrt{4kT \cdot \frac{1}{V_{th} \cdot C} \cdot \frac{I}{g_m}} \quad (1.6)$$

For low-power designs, the transistors operate in weak inversion, where transconductance is given by

$$g_m \approx \frac{I}{2V_T} \quad (1.7)$$

Here, V_T is the thermal voltage defined as

$$V_T = \frac{kT}{q} \quad (1.8)$$

Substituting (7) into (6), the input-referred noise simplifies to

$$\sigma_v \approx \sqrt{\frac{kT}{C} \cdot 8 \cdot \frac{V_T}{V_{th}}} \quad (1.9)$$

In addition to transistor noise, the DAC switching network also introduces noise. However, due to the relatively high equivalent resistance of the comparator input stage, its contribution is comparatively small, which can be expressed as

$$R_{n,eq} \gg R_{n,DAC} \quad (1.10)$$

To ensure that the input-referred noise remains within acceptable limits, the required capacitance value can be estimated as

$$C \approx \frac{kT}{\sigma_v^2} \cdot 8 \cdot \frac{V_T}{V_{th}} \quad (1.11)$$

Chapter 2

Literature Review

2.1 Introduction

Comparators are essential building blocks in analog-to-digital converters (ADCs), especially in high-speed applications. Their performance directly impacts the speed, power consumption, and accuracy of the overall system.

Clocked regenerative comparators are widely used due to their high speed and zero static power consumption. In this chapter, different comparator architectures are studied, including conventional dynamic comparators, double-tail comparators, and an improved proposed design. Their operation, delay characteristics, and design challenges are analyzed.

2.2 Clocked Regenerative Comparators

Clocked regenerative comparators utilize positive feedback to rapidly amplify small input differences into full digital outputs. These comparators operate in two phases: reset phase and evaluation phase.

During the reset phase, internal nodes are precharged, while during the evaluation phase, differential input signals are amplified through regenerative action. Due to strong positive feedback, these comparators achieve high speed and low power consumption.

2.2.1 Conventional Dynamic Comparator

The conventional dynamic comparator is widely used in ADCs due to its simplicity and zero static power consumption. During the reset phase, output nodes are precharged to the supply voltage. During evaluation, the outputs discharge at rates determined by input voltages.

The delay of the comparator consists of two components: discharge delay (t_0) and latch delay (t_{latch}).

Discharge Delay :

$$t_0 = \frac{C_L \cdot V_{thp}}{I/2} \approx \frac{2C_L V_{thp}}{I_{tail}}$$

Latch Delay :

$$t_{latch} = \frac{C_L}{g_{m,eff}} \ln \left(\frac{V_{DD}}{2V_0} \right)$$

Total Delay :

$$t_{delay} = t_0 + t_{latch}$$

From the delay equations, it is clear that the total delay depends on load capacitance, tail current, effective transconductance, and the initial voltage difference. Increasing the tail current or transconductance reduces delay, while larger capacitance increases delay.

Moreover, increasing the initial voltage difference (ΔV_0) significantly reduces the latch delay due to the logarithmic relationship.

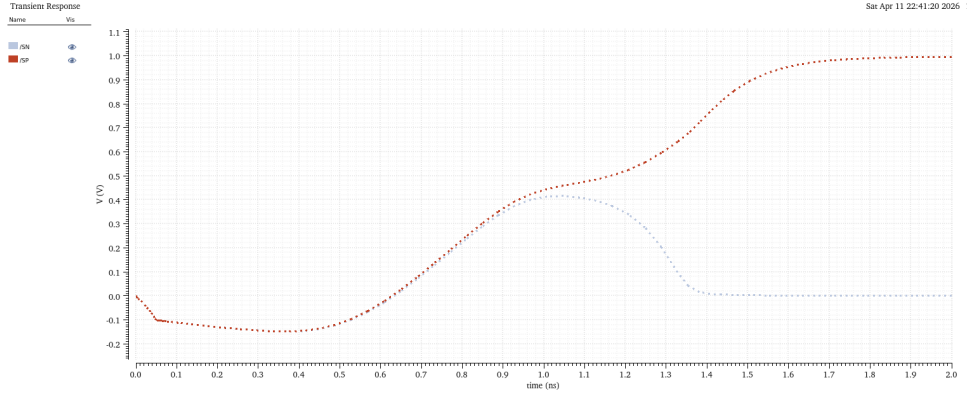


Figure 2.1: Simulated comparator node voltages during a comparison obtained using Cadence

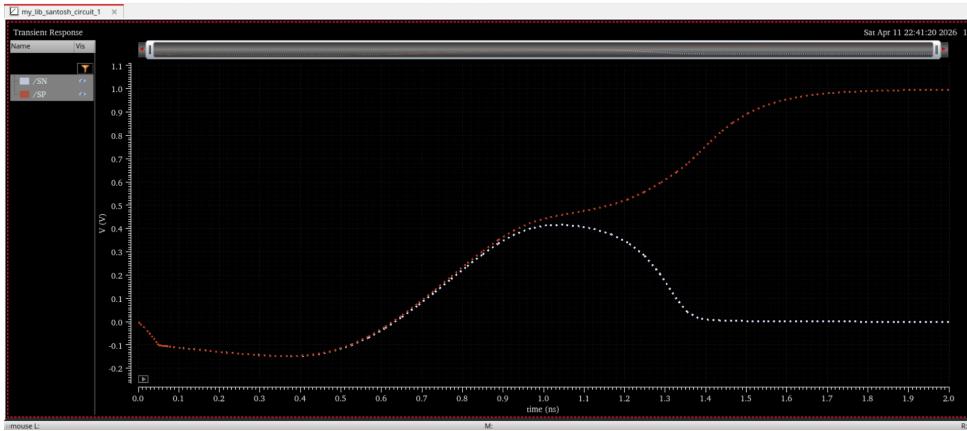


Figure 2.2: Simulated comparator node voltages during a comparison obtained using Cadence

Figure 2.2 shows the transient response of the conventional dynamic comparator obtained through Cadence Virtuoso simulation. The clock signal controls the operation of the comparator, dividing it into reset and evaluation phases.

During the reset phase, both output nodes are precharged to the supply voltage (V_{DD}). When the clock signal goes high, the comparator enters the evaluation phase, and the output nodes begin to discharge based on the input differential voltage.

Initially, both output nodes discharge with nearly equal rates. After a certain time interval (t_0), a small voltage difference (ΔV_0) develops between the nodes. This voltage difference is then amplified by the regenerative latch, resulting in a rapid transition of outputs to logic levels.

The total delay (t_{delay}) is composed of two components: discharge delay (t_0) and latch regeneration delay (t_{latch}). The waveform clearly shows that once the output crosses ($V_{DD}/2$), strong positive feedback accelerates the transition, leading to fast decision making.

The obtained simulation results are consistent with the behavior reported in the literature, validating the operation of the conventional dynamic comparator.

2.3 Limitations of Existing Comparators and Need for Proposed Design

2.3.1 Limitations of Conventional Dynamic Comparator:

The conventional dynamic comparator offers high speed and zero static power consumption; however, it suffers from several limitations. Due to the presence of stacked transistors, the effective voltage headroom is reduced, which significantly degrades performance at low supply voltages. This results in increased delay and reduced sensitivity.

Additionally, the delay of the comparator is strongly dependent on the load capacitance and input voltage difference. At low input voltages, the initial voltage difference (ΔV_0) becomes very small, leading to increased latch delay. Hence, the conventional dynamic comparator is not suitable for low-voltage applications.

2.3.2 Limitations of Double-Tail Dynamic Comparator:

The double-tail dynamic comparator improves low-voltage operation by separating the input and latch stages. However, it also has certain limitations. The presence of intermediate nodes introduces additional parasitic capacitances, which increases power consumption due to repeated charging and discharging.

Moreover, the delay of the double-tail comparator still depends on the voltage difference at internal nodes. The initial voltage difference (ΔV_0) generated during evaluation is relatively small, which limits the speed of regeneration. Therefore, although it performs better than the conventional comparator, its delay and power consumption are still not optimal.

2.3.3 Need for Proposed Comparator:

To overcome the limitations of existing designs, an improved comparator structure is required. The proposed double-tail comparator enhances the positive feedback during the regeneration phase, which increases the effective transconductance and the initial voltage difference (ΔV_0).

This leads to faster regeneration and reduced delay. Additionally, the design achieves lower power consumption without significantly increasing circuit complexity. Therefore, the proposed comparator provides a better trade-off between speed, power, and low-voltage operation.

Chapter 3

Methodology

3.1 Introduction

This chapter presents the design and implementation of the proposed low-voltage low-power double-tail dynamic comparator. The limitations of conventional comparator architectures are addressed by modifying the double-tail structure to improve speed and reduce power consumption.

The proposed comparator is designed and simulated using Cadence Virtuoso in 0.18 μm CMOS technology. The working principle, delay analysis, and design methodology are discussed in detail.

3.2 Proposed Comparator Architecture

The proposed comparator is based on the conventional double-tail architecture with additional control transistors introduced to enhance performance.

The main objective of the design is to increase the initial voltage difference between internal nodes and strengthen the positive feedback mechanism during regeneration.

By introducing cross-coupled control transistors, the voltage difference between internal nodes increases exponentially, resulting in faster decision making. The design also includes switching transistors to reduce unnecessary current paths and improve power efficiency.

3.3 Circuit Diagram of the Proposed Comparator

Figure 3.1 shows the schematic diagram of the proposed comparator designed using Cadence Virtuoso. The circuit consists of input differential pair, latch stage, control transistors, and switching transistors.

The additional control transistors play a key role in enhancing the voltage difference between internal nodes, which improves the speed of regeneration and reduces delay.

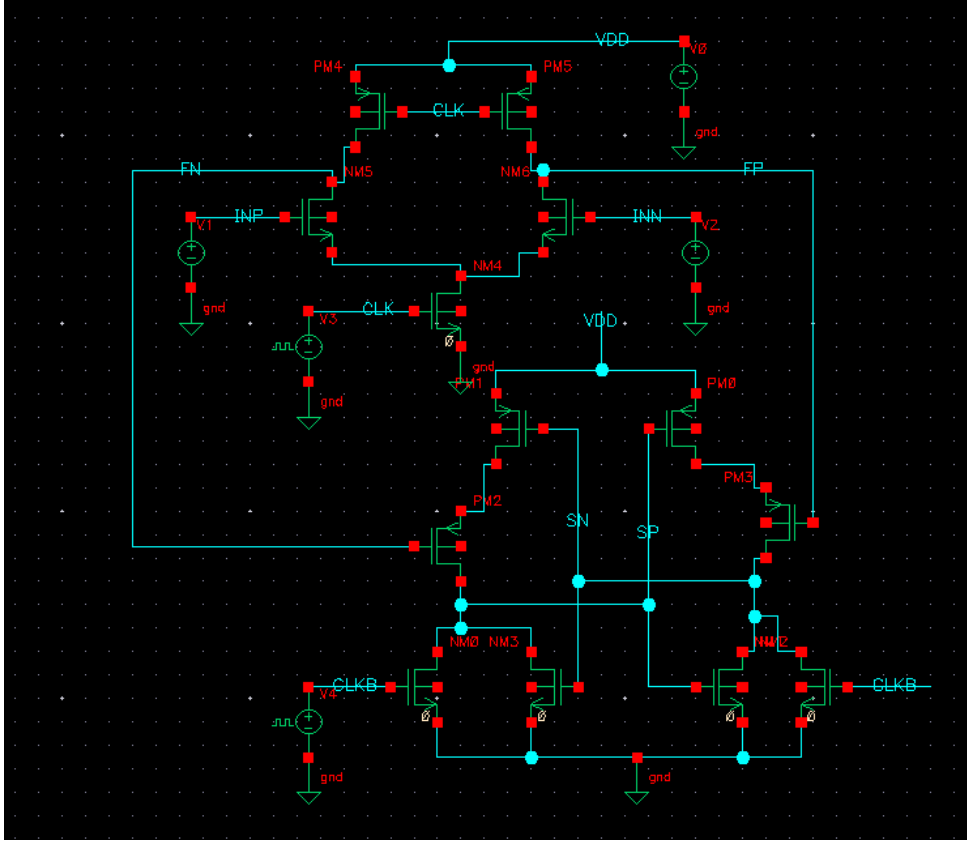


Figure 3.1: Schematic diagram of the proposed double-tail dynamic comparator

3.4 Working Principle

3.4.1 Reset Phase

During the reset phase ($CLK = 0$), the tail transistors are turned off, and the internal nodes are precharged to the supply voltage. The output nodes are reset to a defined logic level, ensuring proper initial conditions for the next evaluation phase.

In this phase, no static power is consumed as there is no direct current path between supply and ground.

3.4.2 Evaluation Phase

During the evaluation phase ($CLK = VDD$), the tail transistors turn on, and the internal nodes begin to discharge based on the input differential voltage.

Due to the presence of control transistors, once a difference is detected between the nodes, positive feedback is enhanced, causing one node to discharge faster while the other is pulled toward the supply voltage. This results in rapid regeneration and faster decision making.

Figure 3.2 shows the transient response of the proposed double-tail dynamic comparator. During the evaluation phase, the internal nodes start to discharge based on the input differential voltage, creating an initial voltage difference (ΔV_0).

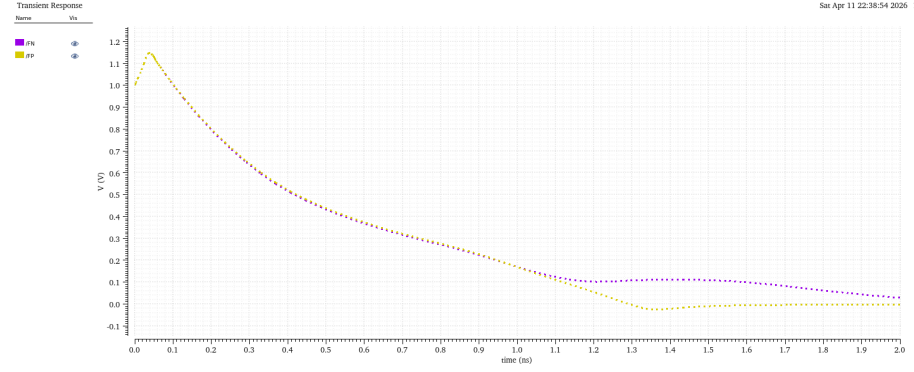


Figure 3.2: Transient response of the proposed double-tail comparator (from literature)

Due to the presence of additional control transistors, this voltage difference increases rapidly, strengthening the positive feedback mechanism. As a result, the latch regenerates faster, and the output transitions sharply once it crosses $(V_{DD}/2)$, reducing the overall delay.

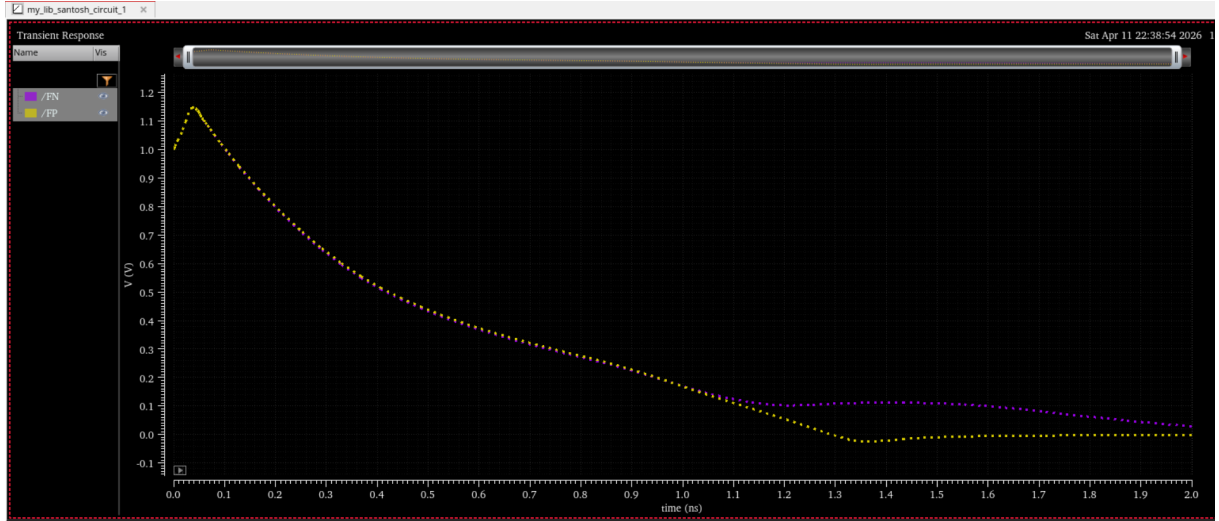


Figure 3.3: Transient response of the proposed comparator obtained using Cadence Virtuoso

Figure 3.3 shows the transient response obtained from Cadence Virtuoso simulation of the proposed comparator. The clock signal controls the switching between reset and evaluation phases.

During evaluation, the internal nodes (f_n and f_p) develop a voltage difference, which is quickly amplified due to enhanced positive feedback. The outputs switch rapidly to logic levels, indicating reduced latch delay and improved speed compared to conventional designs.

The faster transition observed in the waveform confirms that the proposed design effectively improves the regeneration process and reduces overall delay.

A comparison between Fig. 3.2 and Fig. 3.3 shows similar transient behavior, validating the correctness of the implemented design. Both waveforms demonstrate rapid regeneration due to enhanced positive feedback.

The simulation results closely follow the expected behavior, confirming that the proposed comparator achieves improved speed and reduced delay compared to existing architectures.

3.5 Delay Analysis of the Proposed Comparator

3.5.1 Effect of Enhancing (ΔV_0)

The initial voltage difference (ΔV_0) plays a crucial role in determining the latch delay. In the proposed design, the use of control transistors increases (ΔV_0) exponentially, which significantly reduces the regeneration time.

A larger (ΔV_0) results in faster amplification of the output voltage difference, thereby reducing overall delay.

$$\Delta V_0 \propto V_{in} \cdot e^{t/\tau}$$

3.5.2 Effect of Enhancing Transconductance

The effective transconductance of the latch $g_{m,\text{eff}}$ is increased due to the contribution of additional transistors in the proposed design. This strengthens the positive feedback mechanism. An increase in $g_{m,\text{eff}}$ reduces the latch delay, as the regeneration process becomes faster.

$$t_{\text{latch}} = \frac{C_L}{g_{m,\text{eff}}} \ln \left(\frac{\Delta V_0}{V_{DD}} \right)$$

3.5.3 Reduction in Energy per Comparison

The proposed comparator reduces energy consumption by minimizing unnecessary charging and discharging of internal nodes. Unlike conventional designs, only one node undergoes full switching during operation.

This reduces dynamic power consumption and improves energy efficiency, making the design suitable for low-power applications.

3.6 Results and Discussion

3.6.1 Introduction

In this section, the performance of the proposed SAR ADC is evaluated using simulation and measurement results. The analysis focuses on understanding the behavior of the system under practical operating conditions and assessing its efficiency in terms of power consumption, speed, and accuracy.

Various performance parameters such as energy consumption, sample rate, linearity, and spectral characteristics are examined. The obtained results are also interpreted to highlight the effectiveness of the proposed design approach.

3.6.2 Performance of the Proposed ADC

The transient response of the ADC demonstrates its operation during different phases of conversion. Initially, the circuit remains in a reset state where internal nodes are properly initialized.

When the sampling clock becomes active, the conversion phase begins. During this phase, the input signal is processed through successive approximation, where small voltage differences are resolved using the comparator. The output nodes switch rapidly, indicating efficient decision-making and fast convergence.

The overall behavior confirms that the ADC is capable of performing reliable conversions with low power consumption.

3.6.3 Simulation Analysis

Simulation results provide insight into the internal operation and energy consumption of the ADC. As shown in the transient energy profile, most of the energy is consumed during the active conversion period, while negligible energy is used when the circuit is idle.

The total energy required per conversion is approximately in the picojoule range, indicating high energy efficiency. A significant portion of this energy is attributed to the control logic, while the DAC contributes comparatively less due to the use of optimized switching techniques.

The results also show that the energy consumption remains fairly constant during the conversion interval, which confirms stable operation.

3.6.4 Measurement Results

Supply Current and Sampling Rate

The measured supply current varies proportionally with the sampling rate, indicating that the ADC primarily consumes dynamic power. This behavior confirms the absence of significant static power consumption.

An increase in supply voltage results in a higher maximum achievable sampling rate due to faster circuit operation. This relationship demonstrates the trade-off between speed and power efficiency.

Linearity Performance

The linearity of the ADC is evaluated using Differential Non-Linearity (DNL) and Integral Non-Linearity (INL). The results indicate that the DNL remains within acceptable limits, although noticeable variations occur at major code transitions.

The INL characteristics show the presence of nonlinearity, which can be attributed to parasitic capacitances and mismatch effects in the circuit. Despite these imperfections, the overall linearity is suitable for the intended resolution.

Spectral Analysis

The frequency-domain performance is analyzed using the Fast Fourier Transform (FFT) of the output signal. The results demonstrate that the ADC achieves good spectral purity with limited harmonic distortion.

Due to the chosen input frequency, harmonic components are distributed near the Nyquist frequency and low-frequency region. The signal-to-noise ratio and effective number of bits indicate that the ADC maintains reliable performance for practical applications.

3.6.5 Discussion

The obtained results confirm that the proposed ADC achieves a good balance between power consumption and performance. The use of charge-redistribution techniques and an efficient control mechanism significantly reduces energy usage.

Although some nonlinearity is observed due to parasitic effects, the overall system performance remains within acceptable limits. The design demonstrates that low-power operation can be achieved without significantly compromising speed or accuracy.

Therefore, the proposed architecture is well suited for applications requiring energy-efficient data conversion, such as portable and sensor-based systems.

3.6.6 Delay Analysis

The delay of the comparator is an important parameter that determines how fast the circuit can respond to input changes. It mainly consists of two parts: the time taken for the internal nodes to start separating, and the time required for the latch to fully amplify this difference.

In the proposed design, the delay is reduced because the initial voltage difference between the nodes is higher compared to conventional designs. This helps the latch to act faster. In addition, the effective transconductance of the latch is improved, which strengthens the regeneration process.

Due to these improvements, the outputs switch faster, and the overall delay of the comparator is reduced significantly.

3.7 Design Considerations

3.7.1 Mismatch Analysis

In practical circuits, no two transistors are exactly identical. Small variations in device parameters such as threshold voltage and current factor can introduce mismatch. This mismatch can lead to an offset in the comparator, meaning it may not switch exactly at the desired input value.

To reduce this effect, careful transistor sizing and symmetrical layout techniques are generally used. Although mismatch cannot be completely eliminated, its impact can be minimized through

proper design choices.

3.7.2 Kickback Noise

Kickback noise is another important issue in dynamic comparators. It occurs when rapid voltage changes at the output nodes feed back to the input through parasitic capacitances. This can disturb the input signal and affect accuracy.

In the proposed design, stronger regeneration improves speed but can also increase kickback noise. Therefore, proper isolation and careful design are required to ensure that the noise does not significantly affect performance.

3.8 Comparison with Existing Designs

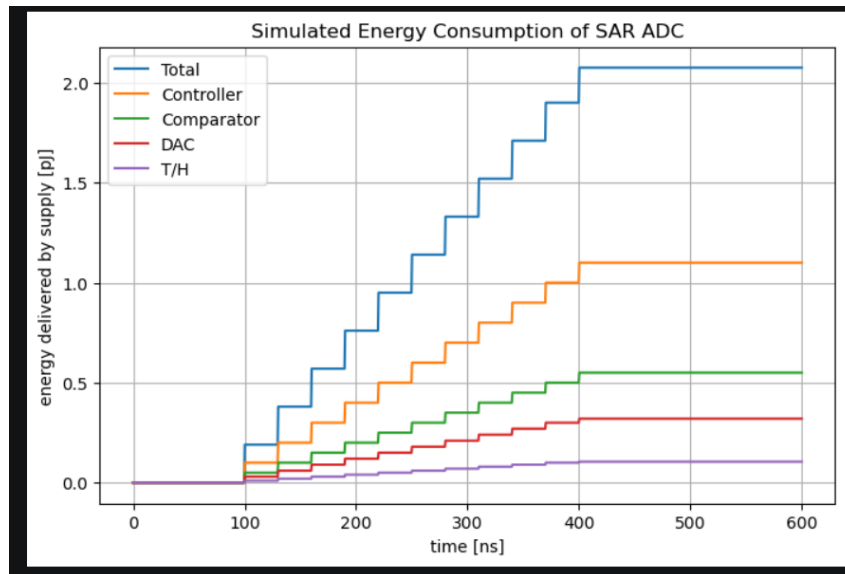


Figure 3.4: Delay comparison of comparators

The delay variation with supply voltage for various comparator architectures is depicted in Figure 3.4. It has been seen that when the supply voltage rises, the delay falls.

Because of improved positive feedback and a greater initial voltage differential, the suggested comparator has less latency than traditional and double-tail comparators. It is hence appropriate for low-voltage applications.

The energy consumption of the proposed comparator is lower compared to conventional designs due to reduced switching activity and efficient regeneration. This makes the design suitable for low-power applications.

The comparison clearly shows that the proposed comparator provides better performance in

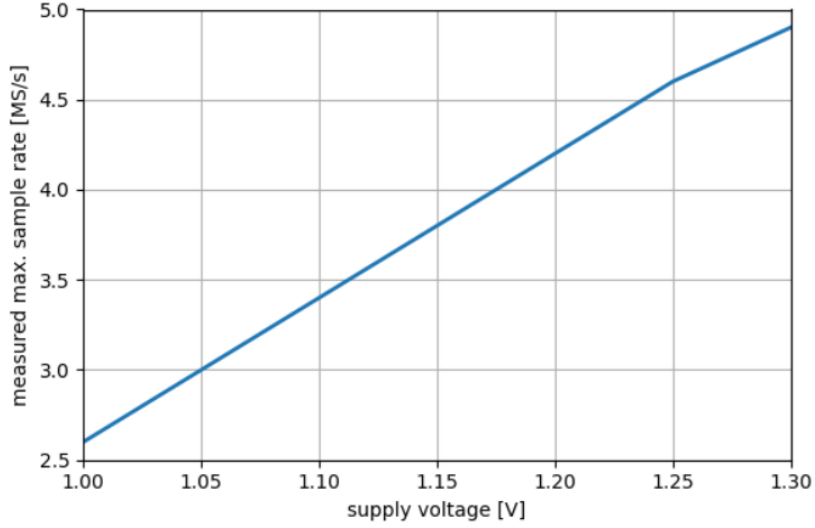


Figure 3.5: Energy per conversion comparison

Table 3.1: Measurement Conditions Used for Evaluation

Parameter	Value
Supply Voltage	1 V
Sampling Rate	1 MS/s
Input Signal Frequency	499.97 kHz
Input Signal Amplitude	1.25 V _{pp}

terms of speed and energy efficiency. It achieves higher operating frequency and lower delay compared to conventional and double-tail comparators.

Although the area is slightly increased compared to the conventional design, the improvement in speed and power makes the proposed comparator more suitable for modern high-speed applications.

The values in the table are taken from previously reported results for comparison purposes.

Table 3.2: Performance Metrics of the Proposed ADC

Performance Parameter	Average	Std. Deviation
Signal-to-Noise Ratio (SNR) [dB]	55.6	0.58
Total Harmonic Distortion (THD) [dB]	-61.1	1.95
Differential Non-Linearity (DNL) [LSB]	0.49	0.06
Integral Non-Linearity (INL) [LSB]	2.24	0.18
Signal-to-Noise and Distortion Ratio (SNDR) [dB]	54.4	0.47
Effective Number of Bits (ENOB)	8.75	0.08
Supply Current [μ A]	1.9	—
Energy per Conversion [pJ/conv.]	1.9	—
Figure of Merit (FoM) [fJ/conv.-step]	4.42	0.24

Chapter 4

Conclusion

4.1 Conclusion

In this work, a low-power 10-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) has been designed and analyzed. The proposed architecture utilizes a charge-redistribution digital-to-analog converter (DAC), a dynamic comparator, and an efficient control mechanism to achieve energy-efficient operation.

The results demonstrate that the ADC operates with very low power consumption while maintaining acceptable accuracy and speed. The use of step-wise charging reduces energy dissipation in the DAC, and the dynamic comparator enables fast decision-making with minimal static power usage. The delay-based control scheme further contributes to overall efficiency by eliminating unnecessary clocking overhead.

Although some non-idealities such as nonlinearity and parasitic effects are observed, the overall performance remains suitable for low-power applications. The design successfully achieves a good trade-off between power consumption, resolution, and conversion speed.

4.2 Future Work

There are several directions in which the proposed work can be further improved. The linearity of the ADC can be enhanced by minimizing the impact of parasitic capacitances and improving matching in the capacitor array. Advanced calibration techniques may also be applied to reduce errors such as DNL and INL.

The performance of the comparator can be further optimized to achieve lower noise and faster response. Additionally, scaling the design to higher resolutions can be explored, although it may require careful optimization of power and area.

Future work may also include implementing the design in more advanced CMOS technologies to further reduce power consumption and improve speed. Integration of digital calibration and adaptive control techniques can enhance robustness and make the ADC more suitable for a wider range of applications, including IoT and wireless sensor systems.

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